TransOilSeg: A Novel SAR Oil Spill Detection Method Addressing Data Limitations and Look-Alike Confusions

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***Abstract*— Marine oil spills pose significant threats to ecosystems and human health, emphasizing the importance of synthetic aperture radar (SAR) images for reliable and all-weather monitoring. However, current methods face two major challenges. The first is data limitations, including insufficient data quantity and noise, such as speckle noise and distortions introduced during preprocessing. The second is look-alike confusions, which pose challenges in distinguishing oil spills from visually similar phenomena. This article introduces *TransOilSeg*, a novel method designed to address these challenges and enhance oil spill detection performance. *TransOilSeg* employs a transfer learning component (TLC) to integrate data from diverse geographical regions and varying quality, learning general features from multisource datasets. By leveraging a gradient aggregation algorithm, the model combines features from limited and noisy SAR oil spill (SOS) datasets, transferring data deficiencies. In addition, the adaptive attention hybrid encoder (AAHE) analyzes contextual features and adapts to varying datasets, enabling the model to effectively distinguish oil spills from look-alike phenomena. Comprehensive evaluations across multiple datasets demonstrate the robust generalization capability of *TransOilSeg*. On the M4D dataset, which includes 1002 training samples, the model achieved a mean intersection over union (mIoU) of 61.38% for oil spill detection and 62.41% for look-alike detection. Furthermore, *TransOilSeg* maintained strong performance when transferred between datasets with varying levels of noise and distortions, demonstrating its adaptability to challenging conditions. These results highlight its potential as a reliable tool for marine oil spill detection and monitoring.**

***Index Terms*— Attention mechanisms, oil spill detection, synthetic aperture radar (SAR), transfer learning.**

I. Introduction

OIL spills have devastating and long-lasting effects on marine ecosystems, typically resulting from oil exploration, transportation, and storage incidents. The rapid detection of oil spills is crucial to minimizing environmental and economic damage. Synthetic aperture radar (SAR) imaging, capable of monitoring day and night under all weather conditions, has become a vital tool for oil spill surveillance [1], [2]. In SAR images, oil spills typically appear as dark regions due to reduced backscattering caused by the damping of sea surface roughness [3]. The irregular and variable sizes of these dark regions necessitate the development of accurate and reliable detection techniques to ensure timely and effective environmental responses.

Traditional SAR-based oil spill detection relies on segmentation techniques that utilize geometric, textural, and polarimetric features [4], [5], [6]. Despite advancements in feature extraction, these methods remain cumbersome and struggle with the high heterogeneity of oil spills [7]. Issues, such as under-segmentation, over-segmentation, limited feature expressiveness, and weak discriminative capabilities of traditional classifiers persist, leaving considerable room for improvement in detection accuracy [8].

The rapid advancement of deep learning has significantly improved oil spill detection by addressing limitations in feature representation and discrimination through hierarchical feature learning [7], [9]. By automatically capturing multiscale spatial details, models such as U-Net and DeepLabv3+ utilize encoder–decoder structures and skip connections to enhance the segmentation accuracy and adapt to the heterogeneous and irregular patterns of oil spills in SAR imagery [10], [11].

Building on these advancements, recent developments have introduced specialized architectures to enhance precision and efficiency in oil spill detection. Mask R-CNN, which leverages a ResNet-101 backbone and a feature pyramid network

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(FPN), effectively distinguishes oil spills in complex SAR imagery [12]. Techniques like multiscale conditional adversarial network (MCAN), employing coarse-to-fine adversarial training across multiple scales, and image filtering combined with CNNs for ensemble modeling, have further improved segmentation performance [13], [14]. CBD-Net enhances multiscale feature extraction and boundary precision in SAR-based oil spill detection through its encoder–decoder structure with squeeze-and-excitation blocks [15]. DGNet incorporates the intrinsic distribution of SAR backscatter values to facilitate efficient learning and accurate segmentation of oil spills, even with limited training data [9].

In addition to segmentation methods, an AI-based approach combines forward scattering models with ANN processing to estimate oil thickness and water-in-oil volume fractions from SAR imagery [16]. Addressing challenges such as elongated oil spill shapes and complex distributions, MIDLN integrates multifeature extraction and semantic complementation to simultaneously localize and segment oil spills [7]. Further advancing detection techniques, SAM-OIL integrates the segment anything model (SAM) with YOLOv8, thereby reducing the dependence on extensive annotations while improving the detection accuracy in SAR imagery [8].

Despite the progress of existing methods, two main challenges remain to be addressed for accurate marine oil spill detection.

Challenge 1 (Limitations of Datasets): The limitations of datasets in oil spill detection primarily manifest in their small quantity, with data reliability and noise further exacerbating the issue [17], [18].

The scarcity of SAR images remains a significant obstacle. High-performance oil spill detection models rely on large and diverse datasets, yet oil spill datasets, such as the M4D and SAR oil spill (SOS) datasets, typically include a limited number of images [19], [20], [21]. In contrast, general-purpose large-scale datasets, such as ImageNet-21K, contain over 14 000 000 images [22]. The limited number of SAR acquisitions restricts the diversity of environmental conditions and spill types. This increases the risk of overfitting and hinders the model's ability to generalize to unseen data [13].

Data reliability and noise further intensify this challenge. SAR images are inherently affected by speckle noise, and preprocessing often introduces additional artifacts such as additive noise and striped distortions [21]. Inconsistent labeling practices and subjective interpretations of spill boundaries further reduce dataset reliability, introducing inaccuracies that hinder model training [23], [24]. Although data augmentation can expand dataset size, it may inadvertently propagate existing errors, complicating training and reducing overall model performance.

Challenge 2 (Confusion With Look-Alikes): SOS detection works by identifying dark regions in radar imagery, which signify potential oil spills due to their reduced backscatter. However, other phenomena, such as biogenic films, algal blooms, upwelling, low wind conditions, and rain cells, can also create dark regions. These look-alike phenomena lead to confusion and complicate accurate oil spill detection [25], [26].

While current deep learning methods perform well in binary classification tasks between oil spills and seawater, most of these methods are not specifically optimized for addressing look-alike issues [14], [15]. To some extent, techniques such as multiscale feature fusion and polarimetric SAR (PolSAR) feature extraction improve differentiation, but they rely heavily on additional data. For instance, PolSAR methods require quad-polarization SAR data, which is limited in availability, whereas most oil spill datasets are single-polarization SAR imagery [27].

Thus, effectively distinguishing oil spills from look-alikes under limited data conditions remains a critical challenge that requires further research and innovative solutions.

To address these challenges, we propose *TransOilSeg*, an innovative framework that integrates transfer learning and adaptive attention mechanisms. Our approach comprises two primary components.

1) *Transfer Learning Component (TLC):* TLC uses large-scale pretraining datasets to learn common features of data, followed by fine-tuning on specific SOS datasets using a gradient aggregation technique. This transfer of shared knowledge enables robust performance even on small or noisy dataset.
2) *Adaptive Attention Hybrid Encoder (AAHE):* AAHE combines CNNs with attention mechanisms to capture both local and global features effectively. It dynamically adapts to varying data characteristics, improving the model's ability to distinguish true oil spills from look-alike phenomena and enhancing classification accuracy across different scenarios.

These components work together to significantly enhance the detection accuracy and generalization capabilities of *TransOilSeg*, ensuring robust performance across diverse and challenging datasets.

We evaluated *TransOilSeg* using the M4D and SOS datasets, which encompass a wide range of oil spill scenarios with varying environmental conditions [11], [15]. Despite the data limitations present in some datasets, *TransOilSeg* consistently outperformed existing methods. Particularly on the mainstream M4D dataset, it achieved a mean intersection over union (mIoU) of 61.38%, surpassing the previous best method by 5.28%, and reached an mIoU of 62.41% for look-alike recognition. These results underscore the model's adaptability and effectiveness, even with data inconsistencies. Cross-dataset evaluations further demonstrated its robust generalization, confirming its potential for practical marine oil spill detection.

The contribution of this article is as follows.

1) *Proposed TransOilSeg Oil Spill Detection Framework:* This study introduces *TransOilSeg*, a novel oil spill detection method utilizing AAHE and transfer learning. The framework learns general features from diverse SOS datasets and optimizes them for specific detection tasks, thereby enhancing the model's generalization capabilities. To support further research, the model is open source and available at: https://github.com/cy7372/TransOilSeg.

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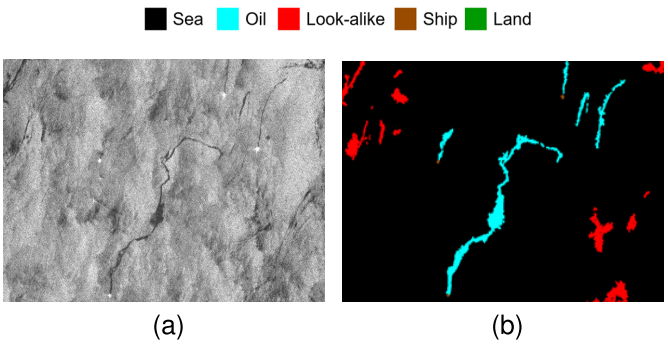


Fig. 1. M4D dataset samples. (a) SAR image showing oil spills and look-alikes. (b) Corresponding labels indicating the identified areas.

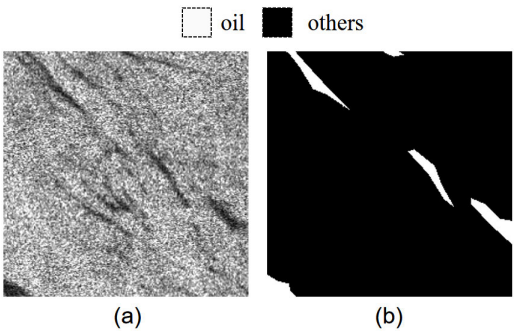


Fig. 2. SOS dataset samples. (a) SAR image showing oil spills and look-alikes. (b) Corresponding labels indicating the identified areas.

2) *Design of the TLC:* We developed TLC, which integrates oil spill datasets from different regions with various annotation characteristics. This approach facilitates broad pretraining and specific fine-tuning for SOS datasets, improving the model's performance in handling small-scale datasets and varying levels of data noise.
3) *Development of AAHE:* AAHE combines attention mechanisms with CNNs to enhance local feature processing and capture extensive contextual information from SAR images. The adaptive layer dynamically adjusts to different dataset characteristics, optimizing the classification process and effectively distinguishing real oil spills from look-alike phenomena.

II. Data Foundation and Motivation Example

A. Dataset Introduction

1) Multimodal Data Fusion and Analytics (M4D) Dataset: The M4D dataset, created by Krestenitis et al. [11], is a multiclass semantic segmentation dataset specifically designed for oil spill detection using SAR images. It includes categories such as oil spills, look-alikes, land, vessels, and sea surface, aiming to advance the development of oil spill detection technologies. Named after the creators' institution, the dataset is supported by geographic coordinates and timestamps provided by the European Maritime Safety Agency's CleanSeaNet service. It contains 1112 SAR images (650 × 1250 pixels), divided into a training set (1002 images) and a testing set (110 images).

The primary challenge of the M4D dataset lies in differentiating oil spills from look-alikes, as both appear as dark patches in SAR images. The coastline and vessel information serve as critical contextual references. This dataset not only facilitates research in oil spill detection but also provides a benchmark for evaluating and comparing detection methods. An illustration of the dataset is shown in Fig. 1.

2) SOS Simulation Dataset: The SOS dataset, proposed by Zhu et al. [15], is a publicly available marine oil spill monitoring dataset. By applying various image enhancement methods, it simulates imaging under diverse conditions and scales, significantly increasing the difficulty of accurate identification. This dataset is extensively used in marine oil spill detection research, particularly for detection from SAR images affected by artificial noise and distortions [21], necessitating high robustness in detection models.

The SOS dataset is created by cropping, rotating, and adding noise to 21 original SAR images and is divided into two subdatasets: Palsar dataset and Sentinel dataset. The Palsar dataset, based on 14 ALOS satellite SAR images from oil spill regions in Mexico, contains 3887 images, with 3101 in the training set and 776 in the testing set. The Sentinel dataset, derived from seven Sentinel-1A satellite SAR images of the Persian Gulf, includes 4193 images, with 3354 in the training set and 839 in the testing set. An illustration of the dataset is shown in Fig. 2.

B. Motivation Example

This section elaborates on the key challenges in marine oil spill detection and demonstrates their impact on spill detection. These challenges primarily include dataset limitations and confusion caused by look-alike phenomena, both of which constrain the accuracy and robustness of SAR-based oil spill detection methods.

1) Dataset Limitations: The modern image analysis and machine learning, particularly deep learning, rely heavily on large datasets to achieve optimal performance [17]. As shown in Table I, datasets such as ImageNet-21K provide extensive training resources, containing 14 900 244 images and occupying 1212 GB of storage. Similarly, MS COCO [28] and VOC2012 [29] offer substantial sample diversity, enabling effective model training.

In contrast, oil spill detection datasets are significantly smaller. The M4D dataset contains only 1112 SAR images with a total size of less than 400 MB, while the SOS-PALSAR and SOS-Sentinel datasets include 3877 and 4193 images, respectively. These SOS datasets, derived from just 14 and seven original SAR images using data augmentation techniques, further highlight the limited volume of real-world data. This stark disparity restricts the diversity of environmental conditions and spill types available for training, posing a significant challenge to building high-performing models.

Beyond quantity, data reliability issues complicate training. Dong et al. [21] identified artificial noise—such as white noise, blurred regions, and distortions introduced during preprocessing—and label noise from misclassified edges or ambiguous pixels. While these issues are secondary to dataset size, they still affect the learning and generalization capabilities of deep learning models.

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TABLE I
SIZE OF COMMONLY USED DATASETS AND OIL SPILL DATASETS

Dataset	Size	Quantity	Purpose
ImageNet-21K	1212GB	14,900,244	Large-scale dataset for training deep learning models.
MS CoCo	25GB	163,957	Large-scale detection, segmentation, captioning dataset.
VOC2012	4GB	11,530	Computer vision dataset for multiple model evaluations.
M4D	387MB	1,112	SAR images for oil spill detection with 5 classes.
SOS-PALSAR	161MB	3,877	Mexico oil spill, based on 14 PALSAR SAR images.
SOS-Sentinel	144MB	4,193	Persian Gulf spill, derived from 7 Sentinel SAR images.

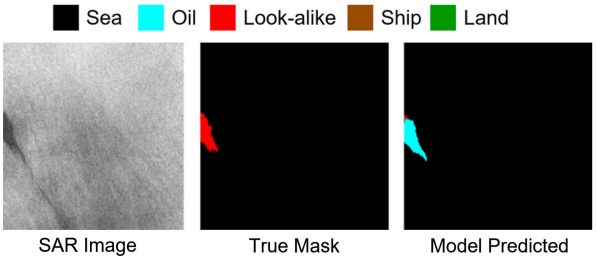


Fig. 3. Example of look-alike problems in SAR imagery for oil spill detection.

In summary, the primary limitation of oil spill detection datasets lies in their small size, which restricts sample diversity and environmental variability. Secondary issues, including artificial and label noise, further exacerbate the challenges, emphasizing the need for robust methodologies to address these constraints.

2) Look-Alike Confusions: The issue of look-alikes in SAR imagery presents a significant challenge for accurately identifying and classifying oil spills, which is critical for environmental monitoring and response. Oil spills on the ocean surface create distinct dark regions in SAR images due to the damping of capillary waves by the oil layer. This damping effect reduces radar backscatter, resulting in a smooth surface that reflects less radar energy back to the sensor, producing characteristic dark regions indicative of oil presence.

However, various natural phenomena can produce similar dark regions in SAR imagery, leading to look-alike issues. Biogenic films, formed by organic substances from microorganisms, create smooth ocean surfaces, reducing radar backscatter and mimicking oil slicks. Algal blooms weaken radar signals, while upwelling smooths the surface with cold, nutrient-rich water, both resembling oil spills. Low wind conditions and convective rain cells also create smooth areas that complicate SAR imagery interpretation. These phenomena significantly hinder the classification of true oil spills, emphasizing the importance of effective differentiation for environmental monitoring [25], [26].

Fig. 3 illustrates this issue. The true mask section shows manual annotations, with blue being oil spills and red being look-alike phenomena. The model predicted section highlights how the U-Net algorithm misidentifies look-alikes as oil spills, underscoring the difficulty of distinguishing real spills from similar phenomena. This highlights the importance of addressing look-alike issues to reduce false positives and improve detection accuracy in SAR-based oil spill detection.

To address these challenges, exploring new detection methods is essential. Such methods must accurately identify genuine oil spill dark regions in SAR images while adapting to limited data resources to enable efficient model training. This research holds both theoretical significance and practical value for advancing oil spill detection techniques.

III. MODEL

To address the challenges described in Section I, this article introduces an oil spill detection model named *TransOilSeg*, as illustrated in Fig. 4. This model leverages multisource datasets by combining large-scale general image collections, such as ImageNet-21K [22], with target SOS datasets such as SOS and M4D. These datasets provide both general knowledge and task-specific features, enabling *TransOilSeg* to learn from diverse sources and improve its oil spill detection performance. *TransOilSeg* comprises two modules: TLC and AAHE. During the training process of *TransOilSeg*, the TLC initially undertakes the initialization operation, learning generic features of oil spill images from datasets with varying spatial domains and annotation features, and then carries out transfer learning to apply the acquired knowledge to the target oil spill tasks; the AAHE learns segmentation features from the given data and, based on convolution, addresses issues resembling oil films through a multihead attention mechanism and adapts to category misalignments across different datasets using an adaptive layer.

A. Transfer Learning Component

The TLC, illustrated in Fig. 5, is designed to address the limitations of SOS datasets, such as limited quantity and data noise. By leveraging knowledge and generic features acquired from large-scale image datasets, TLC transfers this prior knowledge to SOS datasets, thereby improving the model's performance in oil spill segmentation tasks.

The selection of large-scale datasets is guided by three fundamental criteria. Generality ensures that the dataset contains diverse and representative visual features, enabling the model to capture generic image patterns. Scale requires the dataset to be sufficiently large, providing abundant training samples for robust feature extraction and transferability. Finally, accessibility emphasizes that the dataset must be publicly available and easy to obtain, ensuring practicality and cost-effectiveness.

TLC operates through three key steps: initial learning, gradient aggregation, and fine-tuning. Together, these steps

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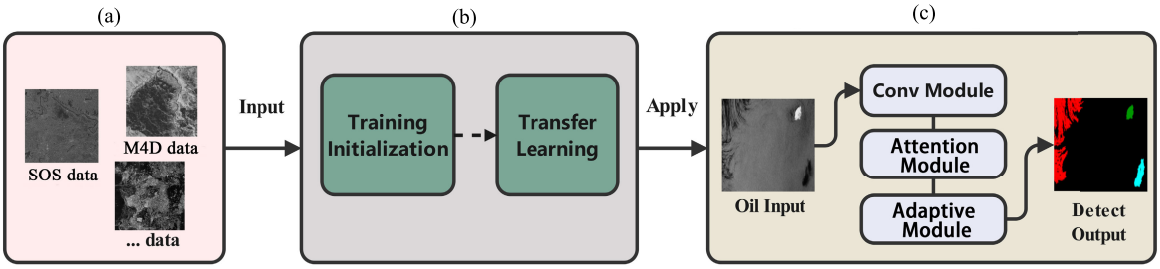


Fig. 4. Framework of *TransOilSeg*. (a) Multisource datasets input, including SOS data and diverse general data, providing the model with rich prior knowledge. (b) TLC (detailed in Fig. 5), which performs initialization and transfer learning to adapt the model to specific oil spill tasks. (c) AAHE (detailed in Fig. 6), which integrates convolutional and multihead attention mechanisms to enhance feature extraction and employs an adaptive layer to adjust to different dataset characteristics, ensuring accurate and robust oil spill detection.

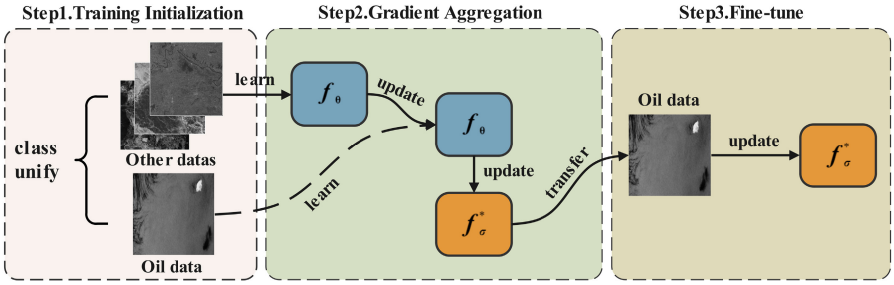


Fig. 5. (Step1) Training initialization. (Step2) The initial model f learns from "other data," updates its knowledge, and then aggregates gradients to transfer this knowledge to the f^* model. (Step3) The f^* model is further fine-tuned for specific oil spill tasks to enhance its performance on the targeted task.

ensure effective knowledge transfer across datasets with different class structures and resolutions, enabling the model to generalize effectively in diverse scenarios.

1) Training Initialization: First, we select a large dataset from the multisource datasets. The TLC uses this dataset to train an initial AAHE, which will be further elaborated in Section III-B.

Subsequently, for each of the remaining N multisource datasets with varying spatial and annotation characteristics, we establish N distinct learning tasks. For each task m, we select a dataset S_m from the multisource datasets and create a unique AAHE f_θ^m, where θ represents the parameters of the AAHE. We initialize these parameters with a random value ϕ^0 and assign it to θ.

Next, data from the oil spill dataset S_m is input into f_θ^m for learning. This involves updating the model parameter θ_m' through one iteration of gradient descent. Each gradient descent follows the formula:

$$\theta_m' \leftarrow \theta_m - \eta \nabla_{\theta_m} l^m\left(f_\theta^m\right) \tag{1}$$

where η represents the learning rate and $l^m(f_\theta^m)$ denotes the learned error.

For each SOS dataset, we update the AAHE f_θ^m in this manner. Based on the updated f_θ^m, we calculate the error of task m using test data, $l^m(f_\theta^m)$ and obtain the gradient θ_m' to update ϕ^0, resulting in ϕ^1. We then select another task n from the learning tasks, perform the same procedure, assign ϕ^1 to the new AAHE f_θ^n, and continue training through the TLC, updating to obtain ϕ^2.

During the training process, the model learns the generic features and related knowledge of SOS images.

2) Gradient Aggregation Phase: During the gradient aggregation phase, the pretrained initial model f_θ^m is transferred to the SOS dataset for further learning. We refer to this method as gradient aggregation.

Instead of the conventional gradient descent method, in this process, we let the model f_θ^m learn from the oil spill dataset, compute the loss $l(f_\theta^m)$, and perform gradient descent. The result from gradient descent is then updated to the new AAHE f_σ^*

$$\sigma' \leftarrow \sigma - \eta \nabla_\theta l(f_\theta). \tag{2}$$

Here, σ represents the model parameters of the AAHE f_σ^* and η is the learning rate.

For subsequent tasks n and AAHE f_θ^n, gradient updating is performed in the same manner.

Through this approach, the TLC uses knowledge and features learned from multisource datasets to update the overall model parameters. This transfers the experience from multisource datasets to the target SOS dataset, aiming to enhance the performance of the oil spill segmentation task.

3) Application of Transfer Component: After the learning in the gradient aggregation phase, we obtained an AAHE f_σ^* that has been optimized on multisource datasets. Next, we need to fine-tune this segmentation model f_σ^* to better adapt to the oil spill task.

We fine-tune f_σ^* by training it on the oil spill dataset and conducting gradient descent

$$\sigma' \leftarrow \sigma - \eta \nabla_\sigma l(f_\sigma). \tag{3}$$

This model, refined through transfer learning, can then be used to segment oil spills in new SOS images. By leveraging the general features and knowledge acquired from the large-scale image dataset, the model demonstrates enhanced performance and generalization capabilities when processing SOS data.

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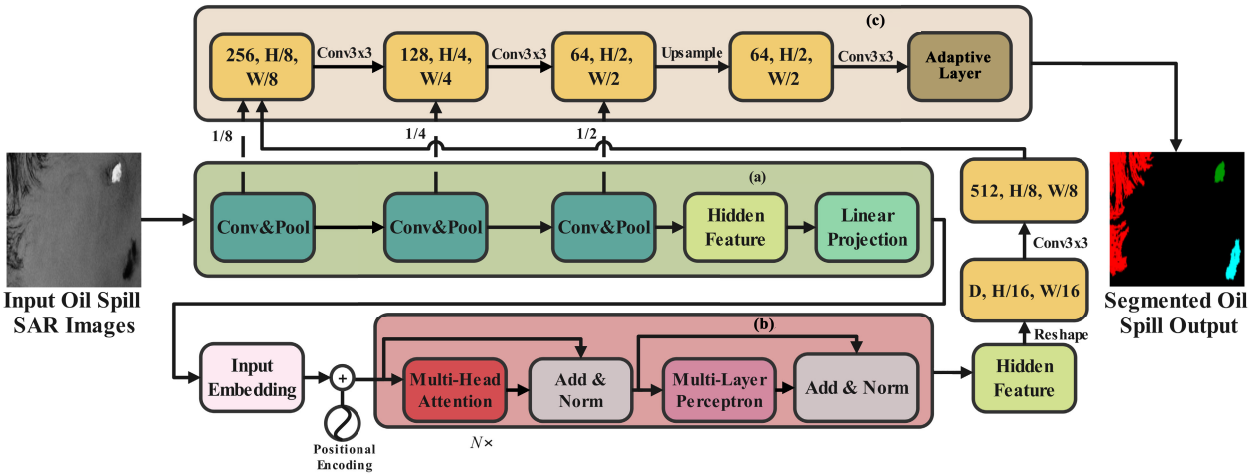


Fig. 6. AAHE. Starting with the input oil spill SAR images, the data progresses through (a) convolutional module, extracting image features via convolutional and pooling layers. These features then enter (b) attention module, which uses multihead attention and MLPs for refinement. (c) Adaptive module allows the component to adjust automatically based on the characteristics of different datasets, optimizing the segmentation process to match the specific output types required. These three modules work together to ensure precise oil spill analysis and detection.

B. Adaptive Attention Hybrid Encoder

This section provides a detailed introduction to the AAHE, as illustrated in Fig. 6. The AAHE consists of three parts: 1) the convolutional module; 2) the attention module; and 3) the adaptive module. The convolutional module is responsible for extracting local features from the image. The attention module learns global features and contextual relationships, while the adaptive module adjusts the final segmentation output based on the characteristics of different datasets. Through the collaborative efforts of these three modules, AAHE is capable of simultaneously considering both the local details and global contextual information of the image, achieving precise oil spill segmentation tasks. This design, which integrates both local and global information, enhances the performance and robustness of the oil spill segmentation model.

1) Convolutional Module:

a) Input image representation: The input SAR image is represented as a matrix X, where $X(i, j)$ denotes the pixel value at position (i, j) in the image.

b) Convolutional layer operation: Initially, a convolutional operation is performed on the input image using a convolutional kernel (weight matrix) W_1 and bias b_1, resulting in the feature map H_1. This operation can be expressed as follows:

$$H_1 = \text{relu}(\text{conv}(X, W_1) + b_1). \tag{4}$$

Here, conv denotes the convolutional operation and relu is an activation function that retains nonnegative feature responses.

c) Pooling layer operation: To reduce the spatial dimensions of the feature map while retaining significant features, a pooling operation is employed to downsample the feature map. A common pooling operation is max pooling, which can be described as follows:

$$H_2 = \text{max_pool}(H_1). \tag{5}$$

Max pooling selects the maximum value from a specific region, thus reducing the size of the feature map.

d) Other convolutional and pooling layers: By stacking multiple convolutional and pooling layers, more advanced feature information is progressively extracted. There can be various combinations of convolutional and pooling layers, and an activation function can be added after each combination for nonlinear transformations.

e) Linear projection operation: A linear transformation is carried out on the input image patch through a learnable weight matrix, mapping it into a latent D-dimensional embedding space. This can be represented as follows:

$$Y = W_2 \times \text{flatten}(H_n) + b_2. \tag{6}$$

Here, flatten unfolds the multidimensional feature map into a D-dimensional vector. W_2 and b_2 are the weights and biases for this layer, respectively. This process involves multiplying each pixel's feature value of the input image patch with the corresponding weight, followed by summing them up. This produces a new feature representation, enabling the convolutional module to learn more.

2) Attention Module:

a) Image serialization after convolutional module: The input SAR image is first reshaped into a flattened 2-D image block sequence

$$\left\{x_p^i \in \mathbb{R}^{P^2C} | i = 1, \ldots, N\right\} \tag{7}$$

where each block is of size $P \times P$ and $N = (HW/P^2)$ denotes the number of image blocks (i.e., input sequence length).

b) Image block embedding: The module uses a trainable linear projection to map the vectorized SAR image blocks x_p into a latent D-dimensional embedding space. To encode block spatial information, the module learns a specific position embedding, which is added to the block embedding to retain

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positional information as expressed

$$z_0 = \begin{bmatrix} x_p^1 E \\ x_p^2 E \\ \vdots \\ x_P^N E \end{bmatrix} + E_{\text{pos}} \tag{8}$$

where $E \in \mathbb{R}^{(P^2C)\times D}$ is the image block embedding projection and the $E_{\text{pos}} \in \mathbb{R}^{N\times D}$ represents the positional embedding.

c) MSA mechanism: In each layer of the attention module, the module applies a multihead self-attention (MSA) mechanism to capture dependencies and contextual information between image blocks. For the oil spill detection task, the self-attention mechanism aids the model in focusing on image blocks related to oil spills, capturing crucial associations between the oil spill and other image elements. The attention encoder consists of L layers of MSA and multilayer perceptron (MLP) blocks. Thus, the output of the lth layer can be expressed as follows:

$$z_l' = \text{MSA}(\text{LN}(z_{l-1})) + z_{l-1} \tag{9}$$

$$z_l = \text{MLP}\big(\text{LN}\big(z_l'\big)\big) + z_l' \tag{10}$$

where $\text{LN}(\cdot)$ denotes the layer normalization operator and z_L is the encoded image representation.

d) Layer normalization: Within each attention layer, the module applies a layer normalization operation after the self-attention mechanism, improving the network's stability and convergence. For the oil spill detection task, the layer normalization helps reduce the variation between different image blocks and better distinguishes the oil spill from other image elements.

3) Adaptive Module:

a) Hybrid encoder: The AAHE employs a hybrid of the convolutional and attention modules as the encoder, responsible for extracting the feature representation of the input image. The CNN acts as the primary feature extractor, generating the input feature map. Image block embedding is applied to 1×1 blocks extracted from the CNN feature map, not directly from the raw image.

b) Cascaded upsampler: The cascaded upsampler (CUP) consists of multiple upsampling steps to decode the latent features. It processes the concatenated feature maps by applying convolutional operations followed by an ReLU activation function, resulting in the final oil spill segmentation outcome. First, the hidden feature sequence output from the attention module, $z_L \in \mathbb{R}^{HW/P^2\times D}$, is reshaped to a form of $H/P \times W/P \times D$, aligning it with the dimension of the feature map. Following that, multiple upsampling blocks are instantiated within the CUP, progressively restoring the resolution of the hidden feature sequence from a low resolution of $H/P \times W/P$ to the full resolution of the original image $H \times W$. Each block sequentially consists of a $2\times$ upsampling operator, a 3×3 convolutional layer, and an ReLU layer.

c) Adaptive layer: In the adaptive module, an adaptive layer is added to adjust the final output based on the characteristics of different datasets. This layer automatically aligns the network's output dimensions with the classification needs of the target task, ensuring that the final output accurately corresponds to the correct classification labels, regardless of the number of categories in the input dataset. This way, the adaptive layer enhances the model's flexibility and accuracy in handling multisource data.

By following the above-described steps, the encoder, which blends the convolutional and attention modules, can simultaneously capture both local and global information about the image. The CUP then restores the image's resolution, ultimately producing the oil spill segmentation mask. This design integrates the capabilities of the convolutional module in handling detailed features with the transformer module's focus on global information, enhancing the accuracy and performance of the oil spill segmentation task.

IV. Experimental Setup

A. Evaluation Metrics

This study utilizes the following metrics for evaluation: precision, recall, $F1$ score, and mIoU, defined as follows.

1) *mIoU:* Calculates the average intersection over union (IoU) for each category. mIoU is widely regarded as the most comprehensive metric and is the most commonly used standard for evaluating the effectiveness of oil spill segmentation [30]

$$\text{mIoU} = \frac{1}{k}\sum_{i=1}^{k}\frac{\text{TP}_i}{\text{TP}_i + \text{FP}_i + \text{FN}_i} \tag{11}$$

where k is the number of categories and TP, FP, and FN are the true positives, false positives, and false negatives for category i, respectively.

2) *Precision:* The ratio of correctly predicted positive observations to the total predicted positives

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}. \tag{12}$$

3) *Recall:* The ratio of correctly predicted positive observations to all observations in actual class

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}. \tag{13}$$

4) *$F1$ Score:* The weighted average of precision and recall

$$F1\ \text{Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}. \tag{14}$$

Due to the existing oil spill detection models utilizing various datasets and evaluation metrics, comparing different models is a complex task that necessitates the design of a universally accepted assessment system. For the two types of datasets used in this study, we have designed corresponding evaluation methods.

For the M4D dataset, mIoU was chosen as the primary evaluation metric, consistent with its widespread use in prior studies [8], [11], [14], ensuring comparability and reproducibility. Since this study focuses on oil spill detection and look-alike, we emphasize the mIoU for the oil category while considering look-alike performance to provide complementary insights. This avoids inflated averages caused by strong performance in ancillary categories like ship or land. To further

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enhance the evaluation, precision, recall, and $F1$ score were included in the baseline comparisons (Tables II and III), offering a deeper understanding of model performance, particularly for challenging categories like look-alikes.

For the SOS dataset, which involves binary classification (sea and oil), we adopted precision, recall, $F1$ score, and mIoU as the primary metrics. These metrics comprehensively evaluate the model's ability to manage false positives, false negatives, and class imbalances, aligning with established practices in the SOS dataset [15], [21].

This evaluation framework ensures consistency with existing practices while capturing the nuanced performance of *TransOilSeg* across datasets and tasks, providing a comprehensive basis for robust comparisons and analysis.

B. Baselines

In this study, we conducted a comprehensive evaluation of the *TransOilSeg* model by comparing it with two categories of benchmark methods. Since many existing oil spill detection models do not make their codes public, limiting their application and evaluation on specific datasets, we prioritized models with publicly available codes or those evaluated on the same datasets with published results. This selection ensures a fair and robust comparison.

Our strategy guarantees the scientific validity of the experimental results and allows for an objective assessment of *TransOilSeg*'s performance. Through this evaluation, we validate the model's effectiveness, practicality, and adaptability in marine oil spill detection. In addition, this underscores the importance of open access to research methods and code, fostering transparency, reproducibility, and progress in the field.

1) Evaluation on the M4D Dataset: For the evaluation on the M4D dataset, we selected a variety of benchmark models, including common image segmentation algorithms and the latest methods for marine oil spill detection. Specifically, we included traditional image segmentation algorithms, such as U-Net and DeepLabv3+, and their performance data were reproduced based on publicly available codes and corresponding datasets. These models were trained and tested by our team to ensure consistency and reproducibility of the results, following the methodology described by Krestenitis et al. [11].

In addition, we incorporated a series of oil spill detection models that have used the M4D dataset for evaluation in the last three years, for a more comprehensive performance comparison. This includes the optimized convolutional method proposed by Luo et al. [31], which enhances the model's spatial resolution capability using dilated convolutions and improves the feature extraction process. In addition, Zhu et al. [15] proposed the CBDNet, a boundary-supervised network designed for SAR image oil spill detection. At the same time, the study by Rousso et al. [14] introduced image processing techniques and integrated models, including variants of U-Net and DeepLabv3+ that utilize contrast stretching and histogram equalization. These methods not only enhance image quality but also reduce generalization errors by combining the advantages of multiple models, thus improving overall detection performance. Finally, the evaluation included the YOLOv8-SAM, SAM-OIL model [8] using the SAM architecture, which focuses on using advanced structures for precise oil spill segmentation, further enhancing the accuracy of oil spill detection.

2) Evaluation on the SOS Dataset: For the SOS dataset evaluation, we included methods, such as U-Net, SegNet, DeepLabV3+, TransUNet, FFDNet-TransUNet, and multi-model ensembles. The evaluation data for these models all came from the work of Dong et al. [21], who optimized their methods for specific challenges of SAR images. In addition, Zhu et al.'s [15] CBDNet was also included; they proposed the SOS dataset and evaluated CBDNet on it.

By comparing these baseline models, we comprehensively assessed the performance and technical advantages of the *TransOilSeg* model, ensuring the scientific nature of the experiments and the reliability of the results. This comparison not only validated the effectiveness of *TransOilSeg* in oil spill detection tasks but also showcased its broad adaptability and potential in practical applications.

C. Training Details

Experiments were conducted on a computer equipped with an Intel Xeon Silver 4210R processor, 128-GB RAM, and 24-GB GPU memory. For the *TransOilSeg* model, appropriate training configurations and optimization strategies were selected.

Two RTX 3090 GPUs were used to accelerate model training, significantly enhancing both efficiency and speed. These high-performance GPUs, with ample memory, effectively handled large-scale computational tasks.

We used ImageNet-21K for TLC to provide the model with a rich foundation of general visual knowledge. Its substantial size and diversity enable the model to capture broad image patterns, enhancing its generalization capabilities. Moreover, ImageNet-21K is publicly available and easily accessible online, aligning with our goal of reducing reliance on large-scale SAR-specific datasets while ensuring the model can be effectively fine-tuned on specialized SAR datasets such as SOS and M4D.

During the pretraining phase, a batch size of 32 was adopted to improve computational efficiency and slightly enhance the model's generalization capability. To optimize convergence speed and performance, the Adam optimizer was chosen, with the learning rate set to $1e^{-4}$, enabling adaptive adjustments to parameter updates during training.

To mitigate overfitting, the dropout rate was set to 0.1, reducing neuron dependencies and improving generalization. A hybrid loss function was employed, comprising 50% Jaccard loss, 25% cross-entropy loss, and 25% Dice loss. This combination accounted for prediction similarity, classification accuracy, and overlap degree, improving oil spill detection performance.

Through these configurations and strategies, the *TransOilSeg* model was effectively trained under the specified hardware environment, achieving commendable experimental outcomes. These choices ensured accuracy, stability, and efficiency throughout the training process.

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TABLE II
MIOU OF *TransOilSeg* AND DIFFERENT BASELINES (M4D DATASET)

Methods	mIoU Results(%)				
	Sea	Ship	Land	Look-alike	Oil
U-Net[2019]	93.90	44.93	92.68	39.55 (+22.86)	53.79 (+7.59)
DeepLabv3+[2019]	96.43	27.63	92.44	55.40 (+7.01)	53.38 (+8.00)
Dilated convolution method[2021]	95.95	44.59	93.05	53.72 (+8.69)	45.18 (+16.20)
CBDNet[2021]	95.52	28.05	92.03	53.13 (+11.28)	51.85 (+9.53)
U-Net with contrast stretch[2022]	92.21	46.25	95.80	53.42 (+8.99)	54.20 (+7.18)
DeepLabv3+ with contrast stretch[2022]	96.07	42.46	95.94	54.51 (+7.90)	54.50 (+6.88)
U-Net with histogram equalization[2022]	96.43	46.26	96.28	55.14 (+7.27)	51.80 (+9.58)
DeepLabv3+ histogram equalization[2022]	96.35	41.37	92.30	57.70 (+4.71)	53.78 (+7.60)
Ensemble model[2022]	96.78	47.28	96.59	58.88 (+3.53)	56.10 (+5.28)
YOLOv8-SAM[2024]	94.34	52.48	87.65	48.15 (+14.26)	41.84 (+19.54)
SAM-OIL[2024]	96.05	52.55	91.81	55.60 (+6.81)	51.60 (+9.78)
TransOilSeg [Our Model]	**97.02**	**33.49**	**94.39**	**62.41**	**61.38**

Note: The columns for *Sea*, *Ship*, and *Land* show only raw values, whereas *Look-alike* and *Oil* also display differences (in parentheses) relative to *TransOilSeg*. Each difference is computed as $\mathrm{TransOilSegValue} - \mathrm{MethodValue}$ (in gray). Positive values indicate that TransOilSeg outperforms the method, while negative values would indicate lower performance.

V. EXPERIMENTAL RESULTS AND ANALYSIS

This section evaluates the performance and generalization ability of *TransOilSeg* in marine oil spill detection through four carefully designed experiments. Each experiment focuses on a specific aspect of the model's capabilities, as detailed in the following sections.

1) *Baseline Comparison and Look-Alike Analysis on the M4D Dataset:* Assess *TransOilSeg*'s ability to detect oil spills and distinguish look-alike phenomena on the M4D dataset, compared with baseline models.
2) *Transfer Performance Evaluation on M4D Dataset:* Evaluate the impact of various transfer learning strategies on *TransOilSeg*'s performance, focusing on the effectiveness of the TLC.
3) *Parameter Experiments (Attention Heads and Patch Sizes):* Examine how attention head configurations and input patch sizes in the AAHE module influence the model's performance.
4) *Comparative Analysis With Baseline Models on SOS Dataset:* Validate *TransOilSeg*'s generalization ability and robustness by applying it to the SOS dataset and comparing it to baseline methods.

The following subsections provide detailed discussions of each experiment and its outcomes.

A. Baseline Comparison and Look-Alike Analysis on the M4D Dataset

1) Motivation: This experiment evaluates the performance of *TransOilSeg* in detecting oil spills and distinguishing them from look-alike phenomena in SAR imagery, compared to existing models. In additiuon, analyzing the model's ability to distinguish oil spills from look-alike phenomena is crucial for validating its practical applicability in real-world scenarios. This dual focus ensures that *TransOilSeg* not only outperforms existing methods in general detection tasks but also excels in resolving specific challenges related to data limitations and target confusion.

2) Method: We conducted experiments on the M4D dataset, which comprises five categories: sea, look-alike, ship, land, and oil. All models were trained and tested on the same datasets to ensure fairness. The mIoU was used as the evaluation metric, consistent with baseline models such as ensemble and SAM-OIL. This approach allows for a robust comparison of model performance, particularly focusing on oil spills and look-alike detection.

3) Results: Table II shows that *TransOilSeg* achieved the highest mIoU in the oil detection category (61.38%), outperforming SAM-OIL (51.60%) and the ensemble model (56.10%). In addition, *TransOilSeg* attained the top mIoU for look-alike objects (62.41%), demonstrating its effectiveness in addressing one of the primary challenges in oil spill detection.

Table III further supports these findings with detailed precision, recall, and $F1$ score metrics. In the oil category, *TransOilSeg* achieved the highest $F1$ Score (74.84%), surpassing other baseline models, reflecting its robustness in identifying oil spills with fewer false positives and false negatives. Similarly, for look-alike objects, *TransOilSeg* achieved an $F1$ score of 76.01%, further validating its capability to differentiate true oil spills from visually similar phenomena.

Although its performance in the ship (33.49%) and land (94.39%) categories was not the highest, Table III shows that the model's metrics for these categories are within the same level as other baseline models, demonstrating no significant differences that would impact the primary focus of this study. These results highlight the robustness of *TransOilSeg*, particularly in the critical categories of oil and look-alike detection,

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TABLE III
PRECISION (P), RECALL (R), AND $F1$ SCORE (F) OF *TransOilSeg* AND DIFFERENT BASELINES (M4D DATASET)

Methods	Sea			Ship			Land			Look-alike			Oil		
	P(%)	R(%)	F(%)	P(%)	R(%)	F(%)	P(%)	R(%)	F(%)	P(%)	R(%)	F(%)	P(%)	R(%)	F(%)
U-Net	97.05	98.75	97.89	48.22	59.06	53.09	90.35	98.94	94.45	79.97	52.74	63.56	73.39	64.24	68.51
DeepLabv3+	98.16	98.16	98.16	28.94	41.74	34.18	91.95	97.77	94.77	78.16	73.08	75.54	66.39	74.48	70.20
Dilated Conv.	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
CBDNet	98.15	97.28	97.71	37.49	52.68	43.81	93.94	97.84	95.85	65.28	74.06	69.39	69.36	67.25	68.29
Ensemble Model	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
YOLOv8-SAM	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
SAM-OIL	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
TransOilSeg(ours)	**98.14**	**98.60**	**98.37**	**50.09**	**47.68**	**48.85**	**95.30**	**98.55**	**96.90**	**80.36**	**72.10**	**76.01**	**74.95**	**74.73**	**74.84**

Note: The table presents raw Precision (P), Recall (R) and F1 Score (F) for the *Sea*, *Ship*, *Land*, *Look-alike*, and *Oil* categories. This complements the mIoU comparisons presented in Table II.

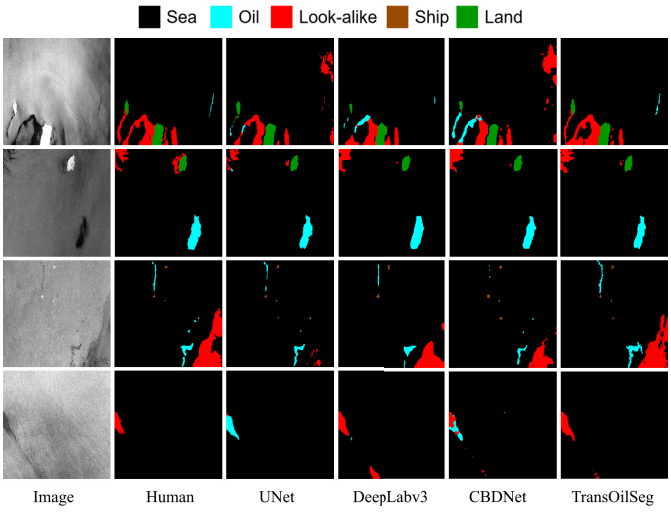


Fig. 7. Comparative segmentation results on the M4D dataset: each row shows test samples with input SAR images, human annotations, and model predictions.

aligning with the study's goal of improving accuracy in these challenging scenarios.

4) Analysis of Look-Alike Recognition: By examining the provided images and their corresponding segmentation results in Fig. 7, it can be observed that while most models misclassify oil spills and look-alike phenomena, *TransOilSeg* successfully achieves more accurate differentiation.

In the first row example, other models consistently misidentify dark regions near the coastline as oil spills. Specifically, although U-Net and DeepLabv3, which rely on global information, can roughly identify dark regions, they still confuse oil spills with look-alikes. Despite CBDNet enhancing contour recognition of dark regions through strengthened contextual information, it similarly fails to effectively distinguish oil spills from look-alikes and misses identifying a dark region on the right side. In contrast, *TransOilSeg* not only precisely delineates the contours of dark regions but also effectively differentiates between oil spills and look-alikes.

In the second row example, the contours recognized by all models are generally similar. However, U-Net and CBDNet erroneously label a small look-alike spot in the dark region at the top left corner as an oil spill. Although both DeepLabv3 and *TransOilSeg* correctly differentiate oil spills from look-alikes in this area, *TransOilSeg* outperforms in terms of boundary and regional precision.

In the third row example, concerning oil spills formed by ship wakes, CBDNet, despite improvements through contextual reinforcement, overly focuses on the lower part of the image and fails to accurately depict the contours of the oil spill. UNet and DeepLabv3 still do not match *TransOilSeg* in global recognition accuracy. As for the dark regions in the lower right corner, lacking reference objects, models can only rely on shape or their own experience to make judgments. *TransOilSeg*, equipped with more prior knowledge, clearly identifies these regions more accurately than other models.

Similarly, in the fourth-row example, when the entire image lacks significant reference objects to aid contextual judgment, the model's assessment of dark regions highly depends on its own experience. In this scenario, *TransOilSeg* successfully classifies the dark region as a look-alike, whereas other models make erroneous classifications.

These qualitative results further validate that *TransOilSeg* effectively combines local and global information, utilizes contextual cues, and leverages the rich experience accumulated through transfer learning, enabling it to perform excellently in complex and dynamic marine environments.

Fig. 8 presents the ambiguous cases, where distinguishing oil spills from look-alike phenomena becomes particularly challenging, indicating that *TransOilSeg* still faces difficulties in certain scenarios.

In the first row example, the entire image contains only a small dark region, lacking other reference information. Moreover, the shape of this region is highly deceptive, making it difficult even for human experts to determine whether it is an oil spill or a look-alike phenomenon-based solely on the image.

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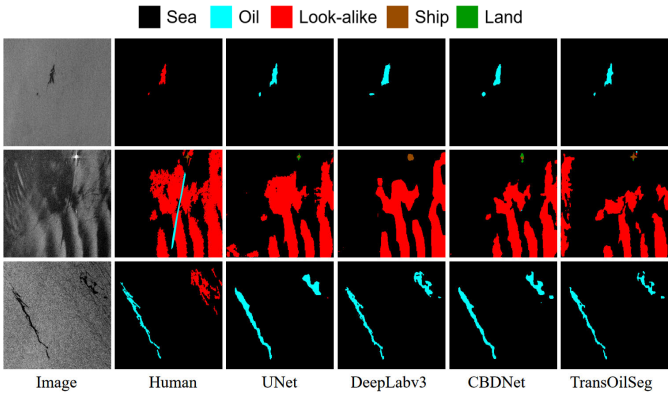


Fig. 8. Ambiguous case analysis on the M4D dataset: examples where *TransOilSeg* struggles to distinguish oil spills from look-alike phenomena. Each row shows test samples with input SAR images, human annotations, and model predictions.

In the second row example, although a ship is present as a reference object and its movement trajectory suggests an oil spill, the oil spill, and look-alike phenomena are mixed and extremely similar in color, making it difficult for the model to accurately distinguish the oil spill based solely on image information.

The third-row example is similar to the first row, with only two black regions in the image, lacking other information. More confusingly, these two black regions are very similar in color and shape, causing the model to classify both as oil spills. However, human experts in the M4D dataset consider the left region as an oil spill and the right region as a look-alike phenomenon.

In the above examples, since *TransOilSeg* primarily relies on image features for judgment, it struggles to accurately distinguish oil spills from look-alikes when contextual information are lacking and the shapes of dark regions closely resemble oil spills, or when oil spills and look-alikes are mixed together, making them difficult to differentiate from the image alone. In such cases, even human experts may temporarily find it challenging to make accurate judgments.

Through the quantitative results in Table II and the qualitative analysis in Fig. 7, it is clear that *TransOilSeg* demonstrates two significant advantages in distinguishing between oil spills and look-alikes. First, the model integrates local textures with global scene information through the AAHE, effectively analyzing contextual information while avoiding over-reliance on local details or global backgrounds, thereby preventing misclassifications. Second, the model accumulates rich prior knowledge through transfer learning, allowing it to accurately distinguish between oil spills and look-alikes even in the absence of reference objects for contextual information, relying instead on its extensive prior knowledge. These two advantages enable *TransOilSeg* to exhibit higher robustness and precision in the complex scenarios depicted in Fig. 7.

However, the confusion cases in Fig. 8 also reveal that *TransOilSeg* still faces difficulties in certain extreme or contextually deprived scenarios. This indicates that although the model performs excellently in most complex situations, distinguishing oil spills from look-alikes remains a challenging task in cases of high similarity or lack of reference objects. These limitations provide directions for future research, such as further enhancing the model's contextual understanding capabilities or introducing more auxiliary information to improve its robustness in extreme scenarios.

In conclusion, by organically integrating local and global information and leveraging the rich prior knowledge obtained through transfer learning, *TransOilSeg* significantly improves the accuracy of distinguishing oil spills from look-alikes, particularly excelling in complex and dynamic marine environments. However, the model still requires further optimization in specific scenarios to address the challenge of identifying oil spills in highly similar or reference-object-lacking conditions.

B. Transfer Performance Evaluation on M4D Dataset

1) Motivation: This experiment aims to evaluate the impact of transfer learning on the performance of *TransOilSeg* in marine oil spill detection. It further assesses the effectiveness of transfer learning strategies in the current context of data limitations.

2) Method: This experiment consists of four sets of tests to evaluate the impact of different data configurations on the performance of the *TransOilSeg* model using the TLC.

1) *Experiment Group 1:* Directly trains *TransOilSeg* using the M4D dataset and tests it on the M4D testing set to assess baseline performance.
2) *Experiment Group 2:* Trains *TransOilSeg* using a mixed dataset of SOS and M4D and tests on the M4D testing set to evaluate the impact of using a hybrid dataset on *TransOilSeg* performance. Since the class categories of SOS (sea and oil) are a subset of M4D's classes, the two datasets were directly combined and trained using M4D's class categories.
3) *Experiment Group 3:* Pretrains *TransOilSeg* on the SOS dataset, then continues training on the M4D dataset and tests on the M4D testing set, assessing the impact of sequential transfer learning. Following a standard fine-tuning approach, the model was first trained using the two-class SOS dataset. Subsequently, the final output layer was modified from 2 to 5 classes, and the model was then trained on the five-class M4D dataset.
4) *Experiment Group 4:* Utilizing the TLC, first train *TransOilSeg* on the SOS dataset, then on the M4D dataset, and finally evaluate it on the M4D testing set. This approach enhances the model's compatibility with knowledge from multiple datasets by automatically adjusting the model's adaptive layers based on the number of classes in different datasets, without the need to modify the output layer.
5) *Experiment Group 5:* Utilizing our transfer component TLC, first train *TransOilSeg* on the M4D dataset, then on the SOS dataset, and finally evaluate it on the M4D testing set. This experiment reverses the training order of Experiment Group 4, thereby assessing the impact of the training sequence on the efficiency of TLC.

3) Results: As demonstrated in Table IV, the *TransOilSeg* model exhibits robust generalization capabilities on the M4D dataset. Experiment Group 1, trained exclusively on M4D,

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TABLE IV
PERFORMANCE EVALUATION OF *TransOilSeg* UNDER DIFFERENT TRANSFER STRATEGIES

Experiment	mIoU(%)				
	Sea	Ship	Land	Look-alike	Oil
Direct Training (M4D Only)	96.43	34.18	92.39	**57.74**	**58.19**
Mixed Training (SOS + M4D)	96.13	31.40	89.17	**51.26**	**55.32**
Fine-tuning (SOS→M4D)	96.54	34.01	92.31	**57.21**	**58.02**
TLC (SOS+M4D→M4D)	96.95	32.59	95.08	**62.45**	**60.55**
TLC (M4D+SOS→M4D)	97.02	33.49	94.39	**62.41**	**61.38**

established a baseline oil spill mIoU of 58.19%. In contrast, Experiment Group 2, which combined SOS and M4D datasets, experienced a decline to 55.32%, likely due to class structure mismatches and differing sensor resolutions. Experiment Group 3, involving sequential transfer learning from SOS to M4D, achieved an mIoU of 58.02%, comparable with the baseline, indicating limited enhancement from pretraining on SOS. Notably, Experiment Groups 4 and 5, which utilized the TLC with different training orders (SOS → M4D and M4D → SOS, respectively), achieved mIoU scores of 60.55% and 61.38%. The minimal difference of less than 1% between these groups suggests that the training sequence has little impact on the efficiency of the TLC, as the TLC's mechanism involves rotating between small datasets (SOS and M4D) each epoch. These findings highlight the effectiveness of the TLC in enhancing *TransOilSeg*'s performance, demonstrating its ability to adaptively transfer knowledge between datasets with varying class structures and image qualities, thereby addressing challenges related to data limitations.

C. Parameter Experiments

In this section, we conduct two sets of parameter experiments to investigate how different configurations affect the performance of *TransOilSeg* on oil spill detection. Specifically, we explore the impact of the number of attention heads in the AAHE and the influence of different patch sizes on segmentation accuracy. All experiments are carried out on the M4D dataset, which encompasses five classes: sea, ship, land, look-alike, and oil.

1) Impact of Attention Heads on AAHE Performance:

a) Motivation: The AAHE integrates attention mechanisms to capture multiscale features. However, the number of attention heads can significantly influence the encoder's capacity. Fewer heads may not adequately capture diverse spatial and feature relationships, while too many heads might introduce redundant or conflicting information. This experiment aims to identify the optimal number of attention heads to balance model complexity and segmentation accuracy.

b) Method: We vary the number of attention heads (Heads) in the AAHE from {1, 2, 4, 6, 8, 10, 12}, training and evaluating *TransOilSeg* under identical conditions. The performance is measured by the mIoU across five categories (sea, ship, land, look-alike, and oil), providing a comprehensive view of segmentation capability.

TABLE V
RESULTS OF *TransOilSeg* WITH DIFFERENT NUMBERS OF ATTENTION HEADS

Heads	mIoU(%)				
	Sea	Ship	Land	Look-alike	Oil
num=1	90.52	0.00	0.00	**0.00**	**0.00**
num=2	91.42	0.00	0.00	**23.51**	**0.00**
num=4	93.81	18.78	88.56	**53.61**	**52.55**
num=6	96.37	16.41	92.86	**56.13**	**57.37**
num=8	97.02	33.49	94.39	**62.41**	**61.38**
num=10	96.61	37.11	92.90	**58.98**	**57.38**
num=12	96.74	36.13	93.31	**60.11**	**56.98**

c) Results: As shown in Table V and illustrated in Fig. 9, increasing the number of attention heads from 1 to 8 leads to substantial improvements in overall mIoU across all categories. Specifically, transitioning from 4 to 8 heads yields significant gains in accurately segmenting complex classes such as oil and look-alike. The trend depicted in Fig. 9 corroborates these findings, indicating a steady rise in segmentation performance with the increasing number of attention heads up to 8. However, beyond eight heads, the performance either plateaus or slightly declines, suggesting that excessive attention heads may lead to overfitting or the capture of redundant features. Therefore, employing eight attention heads strikes an optimal balance between model complexity and segmentation accuracy, providing robust performance without unnecessary computational overhead.

2) Evaluation of Different Patch Sizes:

a) Motivation: In SAR-based oil spill segmentation, patch size plays a crucial role in balancing the detection of fine-grained details (important for look-alike phenomena) and capturing global context (essential for large and irregular oil spills). Smaller patches excel at detecting local textures but may lack spatial context, while larger patches enhance contextual understanding at the cost of increased GPU memory usage and computational load. This experiment systematically evaluates the effect of patch size on segmentation performance and GPU memory consumption, offering insights into the tradeoffs between segmentation accuracy and resource efficiency.

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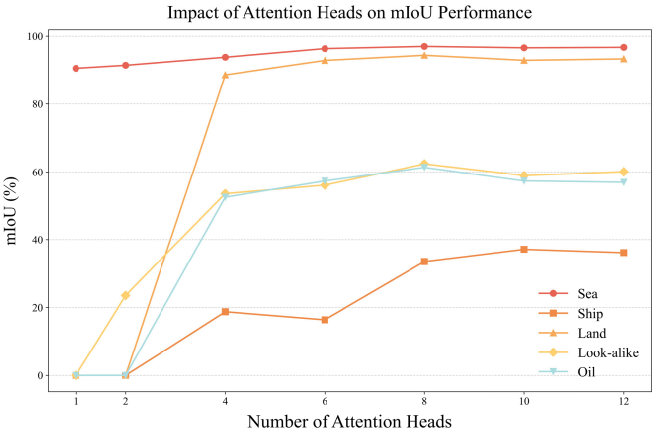


Fig. 9. Impact of attention heads on mIoU performance across different categories.

TABLE VI
RESULTS OF *TransOilSeg* UNDER DIFFERENT PATCH SIZES

Patch Size	mIoU (%)				
	Sea	Ship	Land	Look-alike	Oil
128×128	96.60	24.83	90.24	**59.76**	**54.21**
224×224	97.02	33.49	94.39	**62.41**	**61.38**
256×256	96.88	35.21	94.88	**61.14**	**60.48**
512×512	96.84	42.02	96.53	**62.16**	**61.71**
640×640	96.99	39.42	95.86	**61.78**	**60.60**

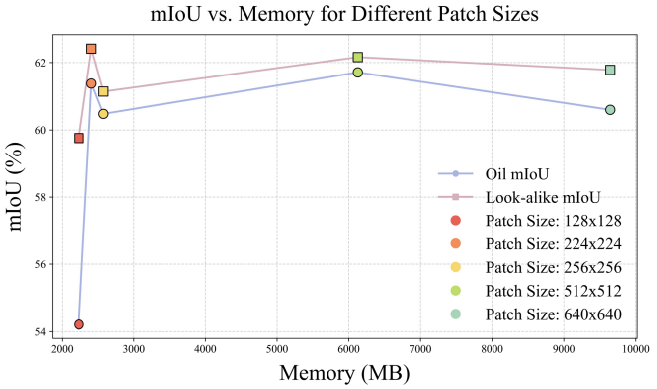


Fig. 10. Oil mIoU versus memory consumption across different patch sizes.

b) Method: We tested five patch sizes: 128 × 128, 224 × 224, 256 × 256, 512 × 512, and 640 × 640. For each configuration, all other parameters, including training and data splits, were held constant to ensure a fair comparison. GPU memory consumption (in megabytes) was recorded during model execution to evaluate computational efficiency alongside segmentation accuracy.

As shown in Fig. 10 and Table VI, the patch size substantially influences segmentation performance and GPU resource consumption. Patch sizes ranging from 224 × 224 to 640 × 640 exhibit comparable segmentation performance, with mIoU for oil remaining between 60.48% and 61.78% and for look-alike between 61.14% and 62.41%. However, GPU memory usage increases significantly as patch size grows, reaching 9648 MB for 640 × 640. This stark rise in resource consumption makes larger patch sizes less practical for resource-constrained applications.

In contrast, the smallest patch size, 128 × 128, shows a marked decline in mIoU (54.21% for oil), primarily due to the loss of global context required for accurately segmenting large and irregular oil spill regions. This result highlights the limitations of smaller patches in balancing local detail detection with global context capture.

While larger patches provide better contextual understanding, their marginal performance gains over 224 × 224 do not justify the significant increase in computational costs. Therefore, 224 × 224 is identified as the optimal patch size, achieving high segmentation accuracy and resource efficiency, making it suitable for real-world SAR-based oil spill detection tasks.

D. Generalization and Robustness Evaluation on the SOS Dataset

1) Motivation: This experiment aims to assess the cross-dataset generalization and robustness of *TransOilSeg* in marine oil spill detection, serving as a complement to Experiment A. Specifically, it evaluates whether the model can maintain high performance when applied to the SOS datasets (SOS-Palsar and SOS-Sentinel), which differ significantly from the M4D dataset in terms of class structure and data distribution. The SOS datasets are binary classification tasks (oil versus others) derived from 21 original images, resulting in over 8000 samples with simpler distribution characteristics. Evaluating *TransOilSeg*'s performance on these datasets verifies its adaptability and effectiveness in diverse real-world scenarios.

2) Method: In this experiment, the *TransOilSeg* model is trained on the M4D dataset and then transferred to the SOS datasets for testing. The model is fine-tuned to accommodate the binary classification nature of SOS-Palsar and SOS-Sentinel. We compare our approach against baseline models, including U-Net, SegNet, DeepLabV3+, TransUNet, CBDNet, FFDNet-TransUNet, and multimodel ensemble, using the same evaluation metrics—precision, recall, $F1$ score, and mIoU—to ensure consistency and fairness.

3) Results: As shown in Table VII and illustrated in Figs. 11 and 12, *TransOilSeg* demonstrates superior performance across all key metrics—precision, recall, $F1$ score, and mIoU—on both SOS-Palsar and SOS-Sentinel subdatasets. On SOS-Palsar, it achieves a precision of 96.94%, recall of 96.52%, $F1$-score of 96.73%, and mIoU of 83.08%. Similarly, on SOS-Sentinel, it achieves a precision of 93.43%, recall of 93.07%, $F1$ score of 93.18%, and mIoU of 82.27%. These results highlight the model's reliability and adaptability in oil spill detection across diverse scenarios.

While *TransOilSeg* achieves slightly lower mIoU values compared to CBDNet (1.15% lower on SOS-Palsar and 0.23% lower on SOS-Sentinel), it significantly outperforms CBDNet in other key metrics. For instance, CBDNet's $F1$ Score on SOS-Palsar is 84.84%, which is 11.89% lower than *TransOilSeg*, and on SOS-Sentinel, it is 87.87% and 5.31% lower. These discrepancies highlight higher false positive and missed

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TABLE VII
PERFORMANCE COMPARISON ON SOS DATASETS (DIFFERENCES RELATIVE TO *TransOilSeg*)

Dataset	Model	Precision(%)	Recall(%)	F1-score(%)	mIoU(%)
SOS-Palsar	UNet[2019]	95.13 (+1.81)	95.66 (+0.86)	95.39 (+1.34)	76.77 (+6.31)
	DeepLabV3+[2019]	94.88 (+2.06)	96.27 (+0.25)	95.57 (+1.16)	77.10 (+5.98)
	SegNet[2019]	94.65 (+2.29)	95.83 (+0.69)	95.24 (+1.49)	75.78 (+7.30)
	TransUNet[2021]	95.67 (+1.27)	96.58 (-0.06)	96.12 (+0.61)	79.77 (+3.31)
	CBDNet[2021]	84.20 (+12.74)	86.75 (+9.77)	84.84 (+11.89)	83.31 (-0.23)
	FFDNet-TransUNet[2023]	96.75 (-0.19)	93.72 (+2.80)	95.21 (+1.52)	77.61 (+5.47)
	Multi-model ensemble[2023]	95.11 (+1.83)	97.28 (-0.76)	96.18 (+0.55)	79.48 (+3.60)
	TransOilSeg(our model)	**96.94**	**96.52**	**96.73**	**83.08**
SOS-Sentinel	UNet[2019]	88.87 (+4.56)	92.26 (+0.81)	90.53 (+2.65)	75.52 (+6.75)
	DeepLabV3+[2019]	88.68 (+4.75)	94.69 (-1.62)	91.59 (+1.59)	77.37 (+4.90)
	SegNet[2019]	90.43 (+3.00)	90.89 (+2.18)	90.66 (+2.52)	76.42 (+5.85)
	TransUNet[2021]	93.41 (+0.02)	88.63 (+4.44)	90.96 (+2.22)	78.06 (+4.21)
	CBDNet[2021]	91.20 (+2.23)	87.32 (+5.75)	87.87 (+5.31)	83.42 (-1.15)
	FFDNet-TransUNet[2023]	91.74 (+1.69)	90.73 (+2.34)	91.23 (+1.95)	77.98 (+4.29)
	Multi-model ensemble[2023]	87.61 (+5.82)	96.79 (-3.72)	91.97 (+1.21)	77.59 (+4.68)
	TransOilSeg(our model)	**93.43**	**93.07**	**93.18**	**82.27**

Note: TransOilSeg serves as the reference. Each metric's difference is computed as $\text{TransOilSegValue} - \text{ModelValue}$ (shown in gray). Positive values indicate that TransOilSeg outperforms the method, while negative values would indicate lower performance.

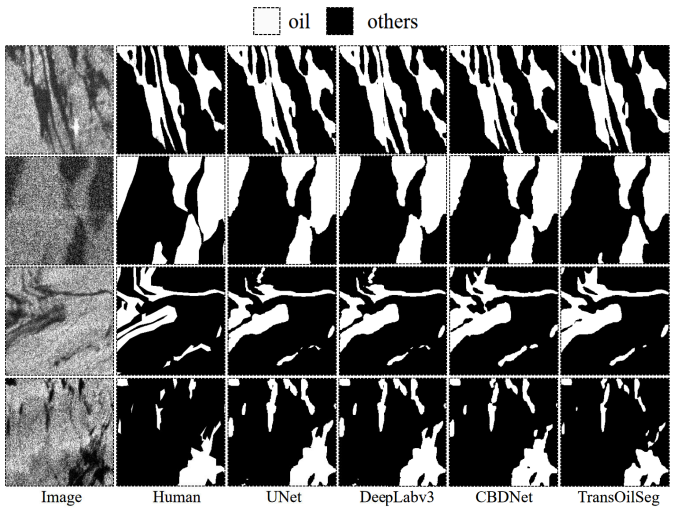


Fig. 11. Segmentation results on SOS-PALSAR dataset: each row shows test samples with input SAR images, human annotations, and model predictions.

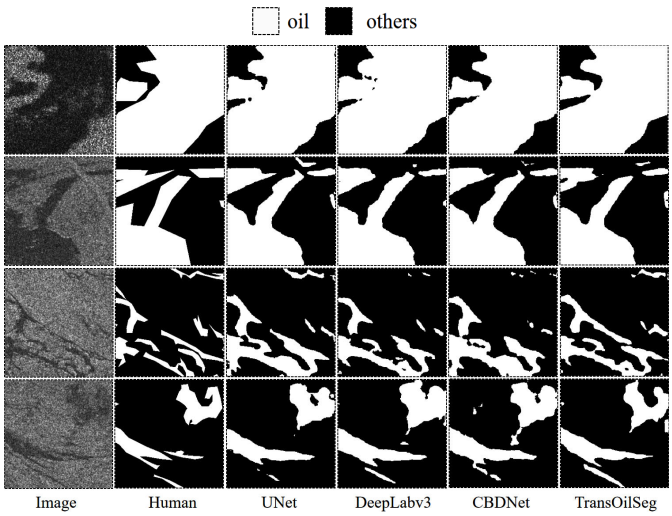


Fig. 12. Segmentation results on SOS-Sentinel dataset: each row shows test samples with input SAR images, human annotations, and model predictions.

detection rates for CBDNet, emphasizing the comprehensive advantage of *TransOilSeg*.

Figs. 11 and 12 visually corroborate these results. On SOS-Palsar, *TransOilSeg* aligns closely with human annotations, effectively delineating oil-contaminated areas while avoiding over-segmentation observed in baseline models. Similarly, on SOS-Sentinel, *TransOilSeg* demonstrates consistent segmentation accuracy, outperforming baseline models in distinguishing oil spills from nonoil regions.

Overall, *TransOilSeg* achieves a superior balance across all evaluation metrics, excelling not only in mIoU but also in precision, recall, and $F1$-score. These results confirm its robustness and generalization capability, making it a dependable tool for real-world oil spill detection in diverse remote sensing environments.

VI. CONCLUSION

In this study, we introduced a novel method, *TransOilSeg*, for marine oil spill detection using SAR images. This method was designed to address two primary challenges in the field: data limitations and look-alike confusion. By incorporating the TLC and AAHE, *TransOilSeg* effectively enhances detection accuracy and robustness across various datasets.

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Our evaluations demonstrated that *TransOilSeg* achieved significant improvements over existing methods. On the mainstream M4D dataset, the model achieved an mIoU of 61.38% for oil spill detection, surpassing the current best method by 5.28%. It also achieved an mIoU of 62.41% for look-alike detection, showcasing its superior ability to differentiate actual oil spills from similar phenomena. Furthermore, the model performed well when transferred between datasets with differing characteristics, indicating its robustness and adaptability.

The contributions of this study are multifaceted. First, the introduction of the TLC allows the model to learn general features from large-scale, diverse datasets while effectively integrating oil spill data from different geographical regions and with varying characteristics, compensating for data deficiencies. Second, the development of the AAHE enhances the model's ability to process local and contextual features, improving its accuracy in complex marine environments. These advancements contribute to a more reliable and effective oil spill detection system.

Despite these promising results, there are still limitations to address. The failure case analysis revealed that *TransOilSeg* struggles in scenarios that include only dark regions without sufficient contextual information. In such cases, the black area could be caused by oil spills or look-alikes such as algae or dead water areas. Due to the lack of contextual cues, *TransOilSeg* faces challenges in distinguishing the true nature of these areas, potentially leading to misclassifications. This limitation underscores the need for auxiliary information or enhanced contextual analysis to improve the model performance in such scenarios.

Future research should focus on expanding and improving the diversity and reliability of SOS datasets and exploring methods to incorporate additional contextual information. Further optimization of the AAHE could enhance the model's ability to distinguish between oil spills and look-alike phenomena in even more challenging conditions.

In conclusion, *TransOilSeg* offers a promising approach to improving marine oil spill detection. By addressing key challenges with novel techniques, this method demonstrates the potential to enhance the accuracy and reliability of SAR-based oil spill surveillance. While these findings highlight its effectiveness, further research is needed to refine and expand its capabilities. Future efforts could build on this work to develop more robust and versatile oil spill detection systems, contributing to improved environmental monitoring and protection.

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